

Gene Set Analyses using Bayesian MAGMA Models

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Gene Set Analysis

Gene set analysis evaluates the **coordinated action of genes or sets of variants** within predefined biological pathways or functional groups.

GWAS identify single genetic variants (SNPs) associated with traits or diseases.

- **Many variants have small individual effects**
→ Use larger datasets or make better use of existing data.
- **Some effects are clustered within functionally related genes or pathways**
→ Use prior information on functional marker groups to improve detection power and interpretation.
- **Some effects are shared across multiple traits**
→ Leverage correlated trait information to enhance detection power and prediction accuracy.

Gene Set Analyses

Many different gene set analysis approaches have been proposed.

- **MAGMA**: Multi-marker Analysis of GenoMic Annotation (Leuww et al 2015)
 - generalized gene set analysis of GWAS data
 - based on a multiple regression model
 - rank gene sets
- **PoPS**: Polygenic Prioritisation Scoring (Weeks et al 2023)
 - leveraging polygenic enrichment of gene features (e.g. gene sets) to predict genes underlying complex diseases
 - based on a multiple regression model
 - rank genes

MAGMA: Linear Model Approach

MAGMA fits a **linear regression model** to test associations between **gene sets** and traits.

1. Aggregate **SNP-level GWAS statistics** into **gene-level statistics**, accounting for LD.
2. Use gene-level statistics as the **response variable**.
3. Represent gene sets as a **predictor matrix**, typically indicating gene membership, but not necessarily limited to binary values.
4. Estimate regression coefficients to assess the **strength of association** between each gene set and the trait.
5. Evaluate significance using **permutation** or **model-based null distributions**.

MAGMA: Limitations

When analyzing thousands of gene sets, several issues arise:

- **Overfitting** – the model may capture noise rather than true signals.
- **Multicollinearity** – many gene sets are correlated due to biological overlap.
- **Multiple testing** – increases false-positive risk.
- **Interpretation difficulty** – hard to disentangle contributions of overlapping sets.

→ Use of **regularization** and **variable selection** to improve model robustness and interpretability.

Additionally, many complex traits are **genetically correlated**, sharing overlapping biological pathways.

→ Incorporating **multi-trait information** in MAGMA can increase detection power and reveal shared genetic mechanisms across traits.

Bayesian MAGMA: Idea

Develop and evaluate a **Bayesian gene-set prioritization approach** using BLR within the MAGMA framework.

- Advantages:
 - Incorporates **regularization** and **variable selection** via **spike-and-slab priors**.
 - Controls false positives and **handles overlapping gene sets**.
 - Provides **posterior inclusion probabilities (PIP)** as evidence of gene set association.
- Flexible framework supporting:
 - **Single- and multi-trait models**
 - **Integration of diverse genomic features**
 - **Modeling of correlated traits** to uncover shared genetic factors.

Bayesian MAGMA: Regression Model

The **Bayesian MAGMA** framework builds on the standard regression model:

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$$

- $\mathbf{Y}$ — vector of **observed outcomes** or **gene-/set-level association measures**
- $\mathbf{X}$ — matrix of **genomic predictors** (e.g., gene membership, functional annotations, or pathway indicators)
- β — vector of **effect sizes** describing how predictors in $\mathbf{X}$ explain variation in $\mathbf{Y}$
- ε — **residual noise** capturing unexplained variation

In the **Bayesian formulation**, each β_j is assigned a **prior distribution** that encodes assumptions about **effect size magnitude, sparsity, or functional grouping**.

These priors enable **regularization, variable selection, and information sharing** across correlated features or biological layers.

Bayesian MAGMA: Multivariate Motivation

- Traditional single-trait analyses may miss associations that are **weak individually but consistent across traits**.
- **Multi-trait Bayesian MAGMA** leverages these correlations by jointly modeling multiple traits to:
 - Increase **power** for detecting gene sets and pathways,
 - Improve **accuracy** of effect estimation, and
 - Reveal **shared biological mechanisms** underlying complex diseases.

Bayesian MAGMA: Multivariate Regression Model

In the **multivariate BLR** model, we model **multiple correlated outcomes** jointly:

$$\mathbf{Y} = \mathbf{XB} + \mathbf{E}$$

- $\mathbf{Y}$: $(n \times T)$ matrix of outcomes
(e.g., association measures for T traits or omic layers)
- $\mathbf{X}$: $(n \times p)$ feature matrix
- $\mathbf{B}$: $(p \times T)$ matrix of effect sizes
- $\mathbf{E}$: $(n \times T)$ residual matrix

Each row of $\mathbf{Y}$ corresponds to an observation or gene, and each column to a trait, phenotype, or molecular layer.

Bayesian MAGMA: Error and Effect Priors

We extend the univariate priors to the multivariate setting:

$$\mathbf{e}_{i.} \sim \mathcal{N}_T(\mathbf{0}, \Sigma_e)$$

$$\mathbf{b}_{j.} \sim \mathcal{N}_T(\mathbf{0}, \Sigma_b)$$

- Σ_e : residual covariance among traits ($\mathbf{e}_{i.}$ is the vector of **residuals** for observation i across all T traits)
- Σ_b : covariance of effect sizes across traits ($\mathbf{b}_{j.}$ is the vector of **effect sizes** for feature j across all T traits)

When the **off-diagonal elements are nonzero**, the model **borrows information across correlated traits**, enabling detection of **pleiotropic effects** and **shared genetic factors**.

When Σ_e and Σ_b are diagonal, the model reduces to T independent univariate BLR models.

Bayesian MAGMA: Indicator Variables

Each feature j may affect multiple outcomes (traits).

We define an indicator vector for **cross-trait activity patterns**:

$$\delta_j = \begin{bmatrix} \delta_{j1} \\ \delta_{j2} \\ \vdots \\ \delta_{jT} \end{bmatrix}, \quad \delta_{jt} = \begin{cases} 1, & \text{if feature } j \text{ affects trait } t, \\ 0, & \text{otherwise.} \end{cases}$$

After Gibbs sampling, the **posterior inclusion probability (PIP)** is estimated as:

$$\widehat{\text{PIP}}_{jt} = \frac{1}{M} \sum_{m=1}^M \mathbb{I}(\delta_{jt}^{(m)} = 1) \approx P(\delta_{jt} = 1 \mid \text{data}),$$

representing the probability that feature j is associated with trait t .

Bayesian MAGMA: Posterior Parameters

In the multivariate setting, we generalize each posterior quantity:

Parameter	Interpretation
$\mathbf{B} = [\beta_{jt}]$	Effect matrix across traits (j : feature, t : trait)
$\mathbf{PIP} = [\text{PIP}_{jt}]$	Posterior inclusion probability matrix (j : feature, t : trait)
Σ_b	Covariance of effects across traits
Σ_e	Residual covariance among traits

These posterior quantities allow us to identify:

- **Shared genetic effects** (pleiotropy)
- **Trait-specific vs. shared signals**
- **Cross-trait enrichment** of biological or functional sets

Bayesian MAGMA. Study Aim and Design

Evaluate a **Bayesian gene-set prioritization approach** using BLR within the MAGMA framework.

Simulation study:

- Assessed model performance under varying gene set characteristics and genetic architectures.
- Used **UK Biobank genetic data** for realistic evaluation.

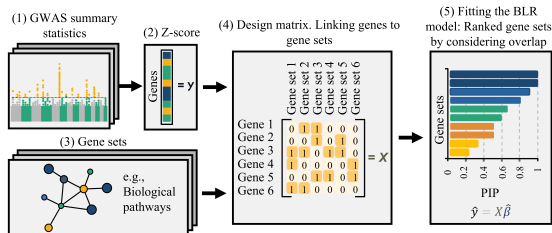
Comparative analysis:

- Benchmarked Bayesian MAGMA against the standard MAGMA approach.

Applications:

- Applied to **nine complex traits** using publicly available GWAS data.
- Developed a **multi-trait BLR model** to integrate GWAS results across traits and uncover shared genetic architecture.

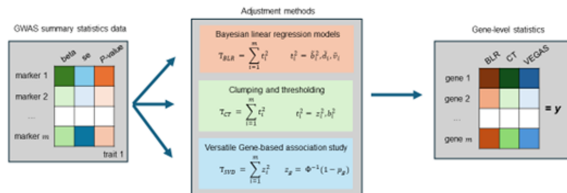
Bayesian MAGMA: Overview



- Fits a Bayesian regression model that allows **regularization** and **variable selection**
- Supports **single- or multi-trait** analyses
- Identifies associated features based on **posterior inclusion probabilities** (PIPs) for the regression effects

Gholipourshahraki et al., 2024

Bayesian MAGMA: Gene-level Statistics



Compute gene-level (or other feature-level) association statistics:

- Account for correlation among marker statistics (i.e., linkage disequilibrium, LD)
- Different LD-adjustment methods (e.g., SVD, clumping and thresholding, BLR)
- The choice of method depends on the quality of the available GWAS summary statistics and LD reference panel

Bayesian MAGMA – KEGG Pathway

- GWAS summary statistics from nine studies (**T2D, CAD, CKD, HTN, BMI, WHR, Hb1Ac, TG, SBP**)
- **Gene sets** are defined by genes linked to **KEGG pathways**.
- Pathways relevant to **diabetes** are associated with **Type 2 Diabetes (T2D)** and correlated traits
- Enables identification of **cross-disease patterns** to better understand **comorbidities**

Gholipourshahraki et al., 2024

Bayesian MAGMA – DGldb

- **Gene sets** are defined by genes linked to the **Anatomical Therapeutic Chemical (ATC)** classification system using the **Drug–Gene Interaction Database (DGldb)**
- **Drug gene sets relevant to diabetes** show associations with **Type 2 Diabetes (T2D)** and related traits
- **Novel drug–gene set associations** may reveal opportunities for **drug repurposing**

Hjelholt et al., 2025

Bayesian MAGMA – Across Ancestries

- **Gene sets** are defined by genes linked to **KEGG pathways**.
- Joint analysis of **T2D** across three ancestries (**EUR**, **EAS**, **SAS**).
- Pathways relevant to **diabetes** show associations with **Type 2 Diabetes (T2D)** across two of the ancestries (**EUR** and **EAS**).
- Comparing these associations helps reveal **ancestry-specific biological mechanisms**.

Summary

Advantages

- Incorporates **regularization** and **variable selection** via *spike-and-slab* priors
- Accounts for **correlated gene sets and traits**, increasing power and reducing false positives
- Provides **posterior inclusion probabilities (PIPs)** as interpretable measures of association strength
- Flexible framework supporting:
 - **Single- and multi-trait models**
 - **Integration of diverse genomic features**
 - **Modeling of correlated traits** to uncover shared genetic architecture

Limitations

- Dependent on the **quality and resolution** of GWAS summary statistics
- Sensitive to **biases and incompleteness** in functional annotations (e.g., tissue specificity, experimental noise, or overrepresentation of well-studied genes)
- **Computationally more demanding** than standard MAGMA, especially for multi-trait analyses

Future Work

- Offers opportunities for **hierarchical model extensions** (e.g., grouping by genomic or functional layers)
- **Multi-omics integration** to capture regulatory complexity across biological layers
- Systematic **comparisons with machine learning and deep learning** methods to evaluate performance and scalability

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Overview of BLR Models used in Gene Set Analyses

Model Type	Feature Integration	Grouping Basis	Prior Structure	What It Captures
Single-component BLR	Combines all biological features in one model	None	One global variance (τ^2)	All features contribute equally; uniform shrinkage
Multiple-component BLR	Integrates all layers but allows heterogeneous contributions	Learned from data	Mixture of variances ($\{\tau_k^2\}$)	Large, small, and null effect classes
Hierarchical BLR	Groups features by biological structure (e.g., genes, pathways)	Defined <i>a priori</i>	Group-specific mixture of variances ($\{\tau_{gk}^2\}$)	Within-group heterogeneity; enrichment and

Learning at Different Levels

Model Level	Key Parameters Learned	What They Represent	How They Are Learned	What We Learn Biologically
Effect sizes	β	Strength and direction of association for each feature	Posterior mean/median given priors and data	Which features drive the outcome
Indicator variables	δ_j (single trait), δ_j (multi-trait)	Whether feature j is active (and for which traits)	Estimated as posterior inclusion probabilities (PIPs)	Which features are relevant, and whether effects are shared or trait-specific
Variance	$\tau^2, \{\tau_k^2\}, \{\tau_{\tau_k}^2\}$	Magnitude of expected	Inferred hierarchically	How strongly different